

TrES-3b

R-0334

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_170415 · created 2026-07-18T05:28:38+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457858.9076 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.073 +/- 0.059
Transit depth (Rp/Rs)^2	0.53 +/- 0.85 [%]
Orbital Inclination (inc)	80.43 +/- 2.86
Airmass coefficient 1 (a1)	0.48 +/- 0.25
Airmass coefficient 2 (a2)	0.58 +/- 0.7
Scatter in the residuals of the lightcurve fit is	52.3 %
Best Comparison Star	None
Optimal Aperture	2.64
Optimal Annulus	7.0
Transit Duration (day)	0.025 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.073 ± 0.059 vs 0.16309 (deviation 1.53σ)
Duration fit vs published	0.025 d vs 0.0567 d (deviation 1.17σ)
Mid-time O–C	0.2 min
Depth SNR	1.2
Residual scatter	52.3 %

Light curve

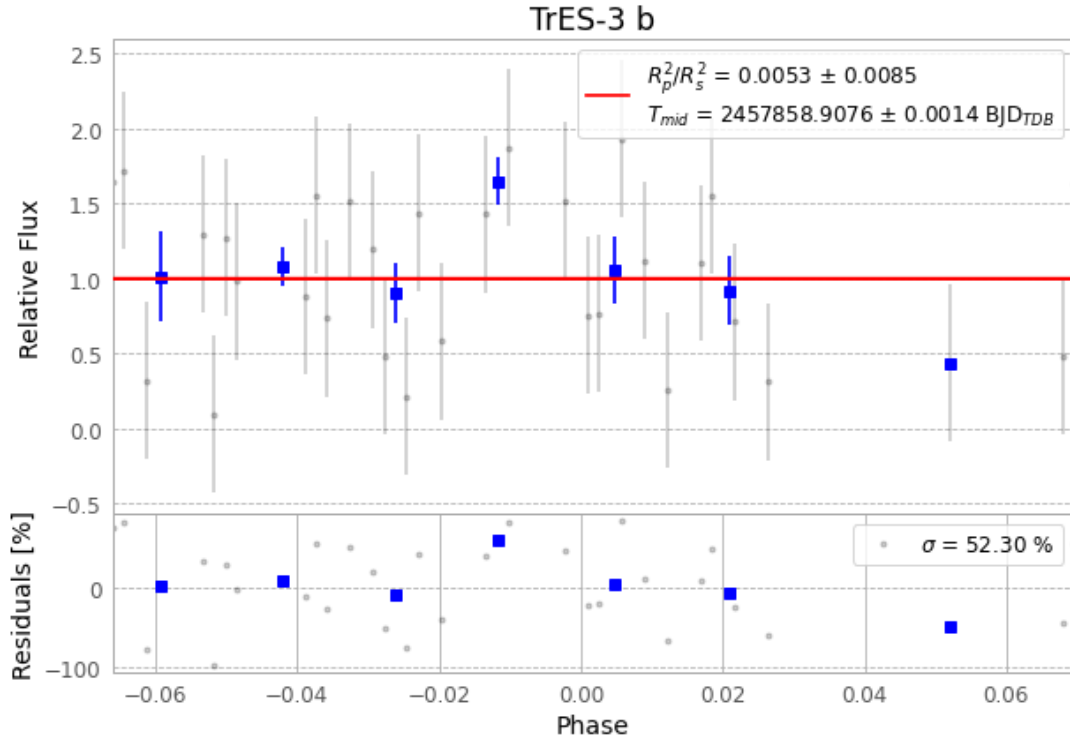


Figure 1 — FinalLightCurve_TrES-3 b_2017-04-15.png (SHA-256 669d05f8beba3cbd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [418, 359], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7862ac832a27c99b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 061fc91

Data provenance

87 FITS frames (57 MB), TRES-3170415073612.FITS → TRES-3170415115412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-170415.log (7e829b3f5bee...), FinalLightCurve_TrES-3 b_2017-04-15.png (669d05f8beba...), FinalLightCurve_TrES-3 b_2017-04-15.csv (952cf9a9e64e...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 05:28Z.